PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013

Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

FULL STACK DEVELOPMENT

AN INTERNSHIP REPORT

Submitted by

MOUNIKA A - 20221CSE0329

Under the guidance of,

Dr. SUDHA P

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

GAIN MORE KNOWLEDGE
REACH GREATER HEIGHTS

PRESIDENCY UNIVERSITY

BENGALURU

APRIL 2026

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Private University Estd. in Karnataka State by Act No. 41 of 2013

Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this report **"Full Stack Development"** is a bonafide work of **MOUNIKA A(20221CSE0329),** who has successfully carried out the internship work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING during 2025-2026**.**

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DECLARATION

I am a student of final year B. Tech in COMPUTER SCIENCE AND ENGINEERING , at Presidency University, Bengaluru, named **Mounika A**, hereby declare that the internship work titled **"Full Stack Development"** has been independently carried out by me and submitted in partial fulfilment for the award of the degree of B. Tech in COMPUTER SCIENCE ENGINEERING during the academic year of 2025-2026. Further, the matter embodied in the internship has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

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INTERNSHIP COMPLETION CERTIFICATE

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SIMPLICITY WITH PRUDENCE

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To,
Mounika A
Full Stack Developer Intern
Prola Tech Private Limited

Subject: Internship Completion Certificate

This is to certify that **Ms. Mounika A** has successfully completed her **Internship in Full Stack Development** at **Prola Tech Private Limited, Bengaluru**, as an Intern from **5th January 2026 to 6th April 2026**.

During her tenure, she demonstrated sincerity, professionalism, and dedication in carrying out her assigned responsibilities. She worked on **Visual Studio Code** and contributed to the development of **Construction Website.**

Her performance throughout the internship period was appreciable, and she contributed effectively to the team.

We wish her all the very best in her future professional endeavours. Thank you for your time and efforts with us.

Sincerely

Charitha PC

Charitha P C
H R
Prola Tech Private Limited

ACKNOWLEDGEMENTS

For completing this internship work, I have received the support and the guidance from many people whom I would like to mention with deep sense of gratitude and indebtedness. I extend my gratitude to our beloved **Chancellor, Pro-Vice Chancellor, and Registrar** for their support and encouragement in completion of the internship.

I would like to sincerely thank my internal guide **Dr. Sudha P, Associate Professor**, Presidency School of Computer Science and Engineering, Presidency University, for her moral support, motivation, timely guidance and encouragement provided to me during the period of internship work.

I am also thankful to **Dr. Asif Mohammed, Professor, Head of the Department, Presidency School of Computer Science and Engineering** Presidency University, for his mentorship and encouragement.

I express my cordial thanks to **Dr. Duraipandian N**, Dean PSCS & PSIS, **Dr. Shakkeera L**, Associate Dean, Presidency School of Computer Science and Engineering and the Management of Presidency University for providing the required facilities and intellectually stimulating environment that aided in the completion of my internship work.

I am grateful to **Dr. Md Ziaur Rahaman,** Internship Coordinator and **Mr. Sakthi S,** Program Internship Coordinator, Presidency School of Computer Science and Engineering, or facilitating problem statements, coordinating reviews, monitoring progress, and providing their valuable support and guidance.

I am also grateful to Teaching and Non-Teaching staff of Presidency School of Computer Science and Engineering and also staff from other departments who have extended their valuable help and cooperation.

MOUNIKA A

ABSTRACT

This report aims to highlight and critically analyze the knowledge and experiences acquired during my internship as a Full Stack Development Intern at Prola Tech, Bangalore. Prola Tech is an IT services firm that provides innovative and scalable digital solutions to their clients. The primary purpose of my internship is to implement my knowledge of various web technologies to perform real-life projects.

During my early days at Prola Tech, I learned the basics of front-end technologies including HTML5, CSS, Bootstrap, among others. For my first task, I had to develop an official website for Prola Tech, keeping into account various industry-standard design guidelines that included designing a clean layout and responsiveness of the web page on various devices.

Furthermore, as part of my practical work during my internship, I worked on another project in which I had to develop a Construction website using various web technologies mentioned above including HTML5, CSS, Bootstrap along with the help of JavaScript language to make the web pages interactive for its users.

Overall, from my internship, I acquired extensive hands-on experience in designing responsive web pages, following coding standards and practices. In addition, it helped me enhance my technical skills, teamwork, and time management.

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ABBREVIATIONS

Abbreviation	Full Form
HTML5	Hyper Text Mark Up Language
CSS	Cascading Style Sheets
IT	Information Technology
SDGs	Sustainable Development Goals
SEO	Search Engine Optimization
UI	User Interface
UX	User Experience
W3C	World Wide Web Consortium
RAM	Random Access Memory
IDE	Integrated Development Environment
ISO	International Organization for Standardization
IOT	Internet Of Things
VS Code	Visual Studio Code
CDN	Content Delivery Network
DPDPA	The Digital Personal Data Protection Act

Chapter 1

INTRODUCTION

Moving from academic learning to actual professional field is a key moment for any engineering student. The reason for the internship is to bridge the gap between theoretical and practical knowledge in computer science due to the dynamic growth in the field of information technologies. Right now, businesses rely heavily on websites to show people what they offer. To connect better with customers and make work smoother inside the company, a well-built website matters a lot. One of those experiences included an internship, which was a valuable learning experience in terms of practical insight regarding web technologies and the construction of professional websites, among other things.

One of the main purposes of taking this internship was to gain firsthand practical experience in the field of Front End Development by applying theory in practice. One of the key aspects in this learning experience involved learning and applying HTML, CSS, Bootstrap, and Basic JavaScript, which are commonly used in web technologies. Firstly, the task involves the design and development of the company's digital brand identity for Prola Tech, as well as an innovative website dedicated to the construction industry, which would showcase its high level of flexibility in different areas of business. In turn, the need for such a project arises due to the current situation where the modern world relies heavily on mobile phones for searching information and other Internet-related tasks. Therefore, traditional websites cannot be used in today's competitive market, where users have access to information via numerous devices. As can be seen from recent statistics and industry trends, there is currently a very high demand for Full Stack Development within the IT sector. This provides a great opportunity for career development for professionals in this area. At the same time, pre-existing technologies in the field of web development used outdated layout methods based on tables and inefficient plugins, which prevented them from success.

1.1 Background

The internship was completed at a company named Prola Tech, which is technology-oriented. It develops software products and provides IT services within a range of fields, including banking, education, retail, and manufacturing sectors. In addition, the organization

applies innovative approaches to develop scalable and efficient applications using existing technologies. The history of this internship is associated with the growing demand for web technologies in the contemporary world. Due to the expansion of online resources and the process of digitalization, companies tend to implement web platforms in order to facilitate access and increase interaction with clients. Recent research indicates that a significant number of users visit websites using smartphones; therefore, responsive web design is critical. In order to gain practical experience, the internship program concentrated on frontend technologies. Thus, the process of learning began with the basics of HTML language that can be applied for creating website architecture. The next step concerned the usage of CSS technologies in the process of page layout design. Finally, JavaScript was applied to create interactive content.

1.2 Statistics Of Internship

In the recent past, there has been a significant rise in the demand for web developers as the use of digital services increases in the world and particularly in India. The Indian Internet and Mobile Association have stated that India has over 700 million internet users with the majority using mobile technology to access websites. The situation has brought the demand for designing responsive and user-friendly web applications in high demand. In cities such as Bangalore, companies are looking for web developers who can create up-to-date websites.

The necessity of this internship is due to the difference between theoretical training and real-world development. Academic training will only give one the theories and principles but not the practical skills in developing a website using various technologies such as HTML, CSS, Bootstrap, and JavaScript among others. Therefore, the internship provided an opportunity to learn how to apply the technologies in creating modern web applications.

1.3 Prior Existing Technologies

Static HTML & CSS were used for web design before the introduction of responsive websites. All designs were created for fixed widths and did not adjust based on the resolution of the screen, making it difficult to use from a mobile device. There were also many cases where a website would look bad in one browser type compared to another and needed to be corrected with numerous repeated lines of code through various "workarounds". The previous way of designing websites had its share of limitations. First, there was no responsive design since desktops were the only focus, so when developing new websites, the new "Mobile First" approach is now required during the web design process. Second, there were no tools such as the Bootstrap Grid System, so all layouts that were created for multiple columns needed to be solved for; consequently, there were many mistakes when creating these layouts. Third, students did not interact with the web at all since websites did not allow for dynamic interactions, which reduced the number of enjoyable experiences from using websites. This project changed outdated practices by utilizing modern HTML5 & Bootstrap thus, providing designers with ranked HTML markup as well as providing the same functionality across multiple platforms. Consequently, this project meets the current standards of flexibility in web design and includes both a clear visual hierarchy and user-friendly navigation options.

1.4 Proposed approach

1.4.1 Aim of Internship:

The primary objective of this internship was to enhance my knowledge about web development and learn more about the designing and implementation process of websites in real-time. I focused on studying HTML, CSS, Bootstrap, and JavaScript which helped me to develop user-friendly websites using front-end technology.

1.4.2 Motivation Behind Internship:

One of the reasons behind pursuing this internship was to connect theory and practice. With the rising demands of web developers in various companies, it is necessary to acquire practical knowledge and skills about web development. The idea to design real-time

websites such as the company website and the construction website provided the motivation to proceed with this task.

1.4.3 Approach:

In the approach adopted to pursue the internship, I first learnt basic HTML for creating a structure of the website. This was further enhanced with CSS to apply styles and layouts. Then I learned Bootstrap for making responsive websites and finally JavaScript to make the website more interactive. The above-listed tools were used for designing two real-time projects: Prola Tech company website and construction website.

1.4.4 Uses of the Project:

This project will have a direct application to designing professional corporate identities and industrial portals. Because of their responsive design, the applications can be used in industries that need access from multiple devices. These industries include Education, Hospitality, and Retail.

1.4.5 Limitations of the Proposed Design:

The proposed design is highly efficient in the front end with regards to responsiveness and consistent UI. However, the current design is more concerned with client-side implementation. In future versions of this model, there should be more attention paid to server-side implementation through complex backend databases and user authentication.

1.5 Objectives

The objectives specified below were set to govern the internship experience at Prola Tech in a professional manner:

1.5.1 Technical and Requirement Analysis:

To conduct requirement analysis of the professional project and chart out its complete Full Stack application workflow, understanding how the frontend and backend of the application work together.

1.5.2 Designing the System:

To design the system in the form of a modular UI, following a Mobile First Strategy and using the Bootstrap Grid system to design responsive companies and construction websites, fitting well across all resolutions.

1.5.3 Implementation:

To use HTML5, CSS3, and Bootstrap technologies to design the website pages, and then implementing required interactions for the construction themed website using JavaScript technology.

1.5.4 System Management:

To effectively conduct the development process through programming and debugging using the Visual Studio Code, while implementing version control using GitHub.

1.5.5 Professional Behaviors:

To get used to an Agile environment of programming and work with Frontend as well as Backend development teams and follow standard professional programming practices.

1.6 SDGs

Sustainable Development Goals (SDGs) by the United Nations act as a plan for a sustainable future. This internship program at Prola Tech not only involved technical knowledge but also aimed to achieve multiple global goals such as digital literacy, innovation, and economic growth that are depicted in Fig 1.1. Specifically, the following are the goals targeted by this internship program:

SDG 4: Quality Education

By providing practical exposure in industry and connecting theoretical education with industrial applications, this internship program enabled the learners to gain specialized technical knowledge in Full Stack Development. The importance of learning new and advanced technology standards helps in meeting the goal of increasing the number of youth with employable skills.

SDG 8: Decent Work and Economic Growth

Developing the latest technological skills related to the IT field like responsive web designing and systems management helps students prepare for the workforce. By creating professional platforms for the banking sector and manufacturing industry, the project will help in increasing economic productivity through innovation and technological advancement.

SDG 9: Industry, Innovation, and Infrastructure

In essence, this internship entailed the creation of digital infrastructure for Prola Tech and for various industries. Incorporating innovative technologies such as Bootstrap technology and using JavaScript to make the platform dynamic makes it easier for small and medium-sized industries to be part of this digital world.

SDG 11: Sustainable Cities and Communities

By developing a specific website for Construction, there is a move towards creating sustainable cities. Information on construction projects would be easily accessible via the web-based digital infrastructure.

Fig 1.1 Sustainable development goals

1.7 Overview of internship report

The present report is a comprehensive account of the Internship performed within the field of Full Stack Web Development at Prola Tech. The first chapter of this report contains an introduction to the internship which includes background, objectives, statistics, previous technologies, and SDG Alignment. The second chapter is a discussion of the literature review and it contains a critical discussion on previous web technologies, principles of responsive design, and web development process. The third chapter contains a discussion of the methodology adopted for this internship and it highlights a step by step discussion on the learning process. Chapter four of the report talks about project management and it discusses the internship timetable, risk assessment and analysis, and resource planning for the internship. The fifth chapter contains a discussion on System Analysis & Design that covers the requirements, architecture, flowcharts, and design process of the internship. Chapter six contains hardware, software, development details, and coding process of this internship. The seventh chapter is a discussion of evaluation and results which covers testing procedures, results, and learning points during the internship. Chapter eight of the report covers the social, legal, ethical, sustainable, and safety aspects of the project.

Chapter 2

LITERATURE REVIEW

Many studies have been conducted on web development with an aim of ensuring enhanced user experience and performance. It is important to note that proper web designing requires the use of semantic HTML5 to improve readability, manageability, and SEO [1], [2], [12]. Modern CSS3 is often used in the development of designs to manipulate flexboxes and grids that enable proper structuring of content [1], [3].

Some researchers have proposed various methods for achieving proper responsive web designing to guarantee good performance in all forms of devices – mobiles, tablets, and desktop. The bootstrap framework has proven to be efficient for such purposes due to its grid structure and pre-developed reusable components [5], [11].

A number of investigations investigate how JavaScript can contribute to enhancing the interactivity and dynamic nature of web applications. Typical areas where JavaScript can be utilized in web applications include form validation, content manipulation, and event handling. These investigations demonstrate that the use of JavaScript can increase the interaction level, although the use of JavaScript may result in certain complexities if not used appropriately. Compatibility with different browsers and performance improvements become common issues investigated in many sources [4].

Further analysis reveals that despite the fact that modern technologies provide highly advanced development tools, certain weaknesses exist in such areas as performance, scalability, and back-end development. In most cases, investigations only focus on the design aspects without considering the entire process of system development. In an effort to cover these weaknesses, a few authors emphasize the combination of appropriate design approaches with proper programming standards and advanced technological solutions [1],[4].

Table 2.1 Summary of Literatures reviewed

Study Focus	Technology/Concept	Key Observation	Limitation	References
Web Page Structure	HTML5	HTML5 provides an excellent structure for the content on a web page.	Design features are limited.	[2], [12]
Styling Techniques	CSS3	CSS3 makes web pages look attractive	It gets complicated for larger projects.	[3]
Responsive Design	Media Queries	The layout adjusts according to the size of the screen.	It needs proper testing.	[1]
UI Frameworks	Bootstrap	Helps speed up the process of development by providing ready-made elements.	Customization is restricted.	[5], [1]
Layout Systems	Flexbox/Grid	Allows making flexible and	Browser compatibility	[3]

		stylish layouts.	issues might occur.	
Interactivity	JavaScript	Provides interactivity to websites	If not coded efficiently, it slows down performance.	[4]
UI/UX Design	Navigation Design	Improves navigation and increases accessibility.	Requires proper planning.	[1]
Web Standards	W3C Guidelines	Helps keep consistency throughout the process of development.	Understanding of web standards is required.	[12]
Fonts and Icons	Google Fonts, Font Awesome	Improves the visual appeal of web pages and provides better readability.	Third-party dependencies.	[6], [7]

Chapter 3

METHODOLOGY

Agile methodology was adopted in our case as the approach that allows learning through iterations, continuous improvement and flexibility in building the product. Agile methodology is popular in software development, since it divides work into smaller chunks and executes it in phases. As my tasks included adopting new technologies and developing websites, the chosen methodology worked perfectly for managing our workflow and tracking the progress.

3.1 Agile Methodology Model

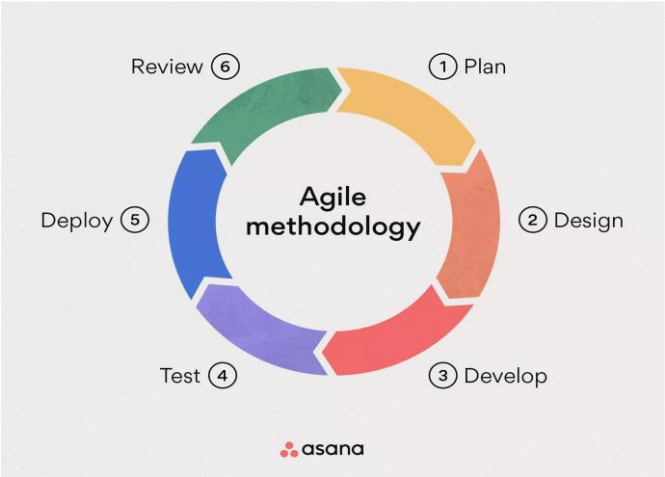

Fig 3.1 Agile methodology model

Fig 3.1 illustrates the model of agile cycle implemented in the internship. As you can see, there were no definite stages but rather iterations, which made the whole process iterative. In this way, both design and functionalities of the websites were improved step-by-step.

3.2 Mapping of Internship Activities to Agile Phases

3.2.1. Planning

In this process, it was made sure that the main objectives of the internship are clear and known. The objective was to gain knowledge about frontend technologies and implement

those skills for designing live websites. Various activities were planned in accordance with the levels of learning from basic HTML and CSS to Bootstrap and JavaScript. It is followed accordingly to complete the required projects.

3.2.2. Designing

While designing websites, planning was done about the layout and structure of the website. In this process, it is decided that how different sections of the website like home, about us, services, contact and other similar pages will be designed. The main emphasis is made on making the website user-friendly and clean.

3.2.3. Development

In this stage, the actual development of the website is done. In order to make the structure of web pages, HTML technology was used. To style and layout web pages, CSS technology was used while Bootstrap technology was used for designing the webpage responsive. Basic functionality like form handling etc. was added through JavaScript.

3.2.4. Testing

It was essential to test the websites created to ensure they worked effectively and efficiently. Testing entailed checking how responsive the websites were when opened from different devices, testing the navigations on various pages, and ensuring that the sites were well-aligned. Any errors detected during testing were sorted out to enhance the website's quality.

3.2.5. Deployment

Upon completion and success in testing, the next step entailed deploying the sites online. The developed projects were arranged and deployed using versioning control platforms such as GitHub.

3.2.6. Review

The last step entailed reviewing the entire process to determine whether the developed projects were as per the required standards. Any feedback received regarding the websites was analyzed and used to make changes in design, responsiveness, and functionality.

Chapter 4

PROJECT MANAGEMENT

Project Management plays a crucial part in completing the internship activities on a systematic basis. During my internship period, I had ensured proper planning, scheduling and monitoring to do assigned tasks in an efficient manner. It was done by dividing the task into various phases such as learning, designing, developing, testing and documenting to manage time efficiently and complete the tasks.

4.1 Internship Timeline

During this internship, there were 12 weeks (3 months) duration. All activities were planned to learn gradually and implement concepts properly related to web development.

- Week1:- Introduction to internship, Understanding of company & project requirement
- Week 2:- HTML basics and creation of web pages
- Week 3:- Advance HTML & introduction to CSS
- Week 4:- Learnings in CSS (Layouts, Styling, Positioning)
- Week 5:- Responsive design and media query
- Week 6:- Bootstrap framework and grid system learnings
- Week 7:- Design & development of company website of Prola Tech
- Week 8:- Employing Bootstrap components in website along with responsive nature
- Week 9:- JavaScript basics and script development
- Week 10:- Site designing and development for construction
- Week 11:- Testing & Debugging

- Week 12:- Documentation and Submission of project

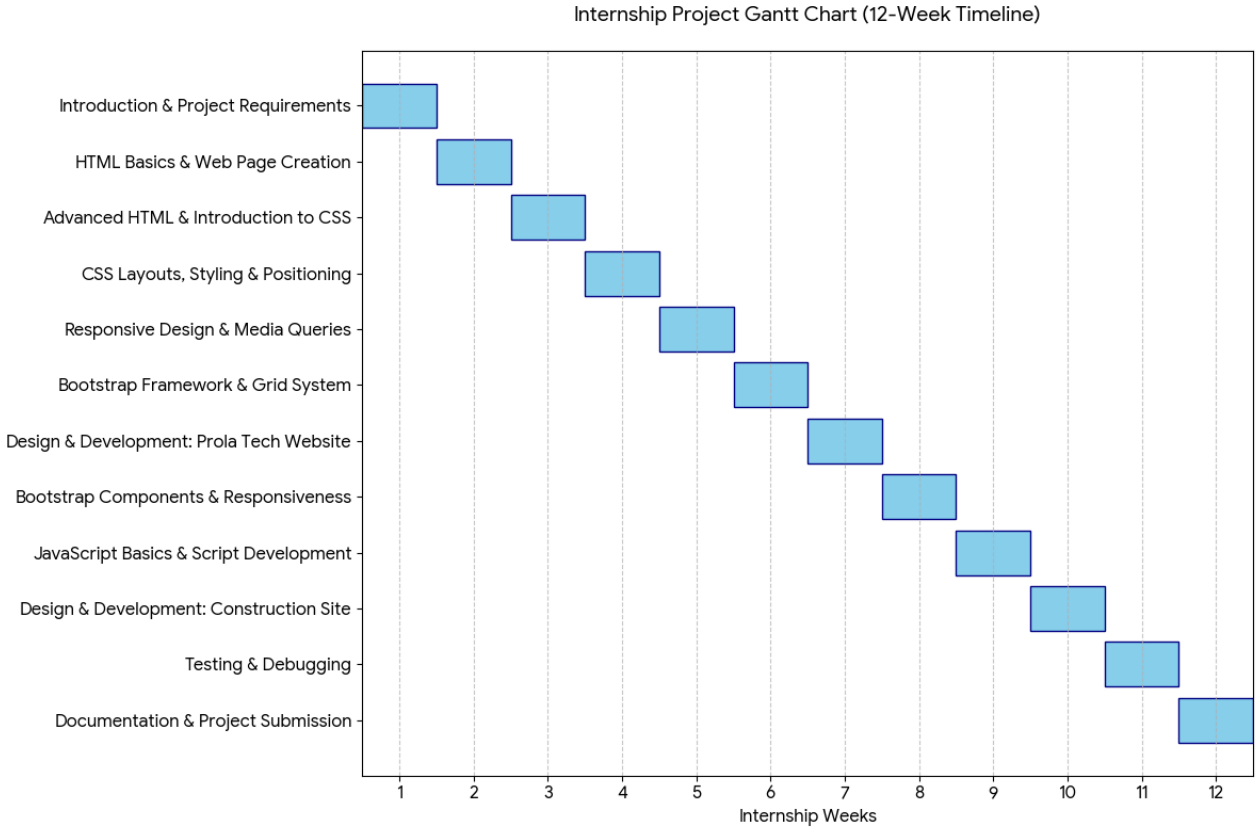

Fig 4.1 Internship Timeline (Gantt Chart)

Figure 4.1 represents the timeline of the internship in three months. The timeline demonstrates how the process started from understanding the fundamental concepts and proceeded to designing complete websites until testing and documentation. This organized timeline ensured efficient use of time and task completion.

4.2 Risk Analysis

There were different kinds of risks that emerged during the internship period, which included technical and professional risks that should be managed carefully for the success of the project despite the steep learning curve for the technology.

- Technical Learning Curve: The process of learning coding standards and complex CSS designs was risky for the rapidity of the development process.

- Mitigation: Through regular discussions among the group members and instructions by the Project Manager, these standards could be learned fast.
- Cross-Browser Compatibility: Custom CSS designs might not display properly in all web browsers.
- Risk Mitigation: Adopted the Bootstrap Framework for its built-in cross-browser compatibility.

4.3 Internship Budget

As this is a software-related internship related to frontend development, there are only "Zero-Cost" open source products along with existing hardware devices included in the budget. Thus, this is a very sustainable and low-cost project.

Table 4.1 Resource Budget for Internship

Category	**Resource**	**Cost**
Hardware	Laptop(min. 4GB RAM)	Existing Resource
Development IDE	Visual Studio Code	0 (Open Source)
Frameworks	Bootstrap 5	0 (Open Source)
Version Control	GitHub	0 (Open Source)
Testing Tools	Google Chrome	0 (Open Source)
Internet	Institutional access	0 (Open Source)
Total Budget		0 (No Additional Cost)

Chapter 5

ANALYSIS AND DESIGN

This chapter provides an analysis of structural and functional needs of the web-based systems implemented during my internship experience at Prola Tech. It is important to differentiate between functional and non-functional requirements in order to make sure that our final solution will address the needs of both, our client and end-user.

5.1 Requirements

In order to create professional and scalable digital products I defined the following types of project requirements. This approach allows us to clearly define the scope of functionality that we have to implement(Functional) and the qualities that our solutions have to possess(Non-Functional).

5.1.1 Functional Requirements

1. Development of Web Pages

The system should enable the creation of HTML web pages that follow a structured format. Each web page should have components like the header, navigation bar, content section, and footer for proper formatting of information.

2. Website Styling and Designing

The system should permit website styling with CSS. The system should enable users to use colors, font styles, spacing, and alignment for a better look of the website.

3. Responsive Design

Our system has to offer an adaptive layout for mobile, tablets and desktop screens, by applying Bootstrap grids to adjust automatically to various screen resolutions.

4. Navigational Hierarchy

Users have to navigate through the website easily, by applying hierarchical design of pages,

including Home, Services, Portfolio and Contact parts of the site.

5. Interactivity

In order to implement interactive elements, we will use JavaScript programming language on the Construction Website.

6. Content Display

In order to organize structured content properly, we will use semantic tags offered by HTML5 language – header, section and footer.

5.1.2 Non-Functional Requirements

1. Performance

The website pages have to open fast irrespective of the speed of the network. Minimized CSS and efficient coding are essential factors in reducing any delay on the user's side.

2. Usability

The site must be straightforward to use. Even users who do not possess any knowledge of computer programming will be able to understand how it works easily.

3. Reliability

The websites must display their content properly each time the user visits them. Testing on Chrome and Firefox browsers would ensure that the layouts work across all platforms.

4. Maintainability

Through the utilization of semantic HTML5 and Bootstrap, the codes are arranged in such a manner that makes them easily maintainable.

5.2 Block Diagram

Fig 5.1 Functional Block Diagram of Web Application

1. User Layer

This layer constitutes the entry point where the user will be interacting with the website. The application will be accessed through the browser where users make different requests to access various web pages.

2. Browser Layer

The browser constitutes an interface between the system and the user. This layer uses HTML, CSS, and JavaScript languages to parse and display output.

3. HTML Layer (Structure)

This layer is responsible for ensuring that the structure of the web page is developed. It incorporates elements such as headings, paragraphs, images, and links to define the content of the web page.

4. CSS Layer (Styling)

CSS enhances the aesthetics of the website by incorporating colors, fonts, spaces, layout designs, and other features into the web page.

5. Bootstrap Layer (Responsiveness)

Bootstrap makes the website responsive through the use of a grid design and pre-built components for the purpose of responsiveness.

6. JavaScript Layer (Interactivity)

JavaScript introduces interactivity in the web page. It is able to respond to user requests through buttons clicks, form validations, and content updates among other features.

7. Output Layer

The final output will constitute a fully-functional responsive web page that can perform all the functions intended in the development process.

5.3 System Flow Chart

The system flowchart explains the process that takes place during the development and functioning of the web application. This visual tool describes how the process of using the web application works, from user interaction through to the end result.

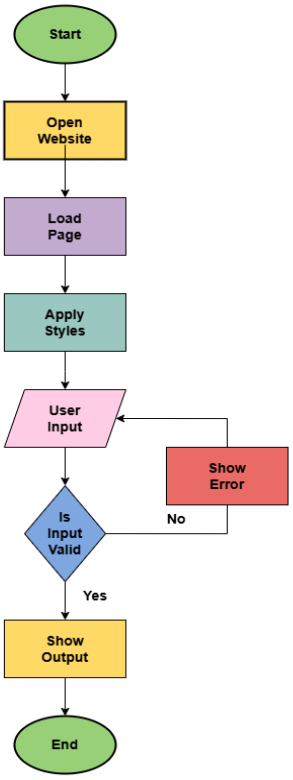

Fig 5.2 System Flow Chart of Web Application

Figure 5.3 presents the flowchart for the web application. Initially, the web application is accessed by the user by opening the webpage. The system uses HTML to construct the page, while CSS adds styling and Bootstrap creates responsiveness in the interface design. Afterward, JavaScript is loaded, allowing the user to interact with the system. If an incorrect entry is made, the system will inform the user and prompt them to enter their input again. Once a correct input is made, the system handles the request and displays its result.

5.4 Choosing devices

In the current project, there is no use of any hardware device including microcontrollers, sensors or actuators. The project is purely about web development, hence there are some computing devices that were chosen to ease the implementation and execution of the project.

The devices that were chosen include the laptop or desktop computer. This system had to have sufficient processing power as well as RAM to facilitate the efficient coding and testing of web applications. It also had to provide sufficient storage space to store coding files and other necessary information.

The tools that were used for software development include code editor and browser applications. The code editor tool facilitated the efficient writing and management of the HTML, CSS, and JavaScript codes while the browser applications enabled testing and validation of websites developed.

The choice of devices and software tools is dependent on simplicity and efficiency since the project is mainly concerned with web development. In addition, the project does not involve any special requirements that would require the inclusion of additional devices.

5.5 Designing units

The system was developed in segments to make it more organized and simpler to develop. These segments include specific components of the application, which may include the structure segment (HTML), styling segment (CSS), responsiveness segment (Bootstrap), and interactivity segment (JavaScript). Each segment underwent independent development and testing before integration into the system. The advantage of developing the system in segments is that it makes it simpler to debug and correct errors. It also ensures that any

changes made to one segment do not affect the whole application.

5.6 Standards

In any professional technical project, standards are essential specifications ensuring proper communication between different systems and devices. While interning at Prola Tech, following certain standards was required to make sure the websites created by me were professional and in compliance with international rules.

5.6.1 Web and Data Standards

To ensure that the websites of Prola Tech and Construction companies would be available via any browser and device, the following standards have been followed for web content:

- W3C HTML5 & CSS3: This standard is used for defining a structure and style of websites so that browsers such as Chrome and Firefox could properly interpret the websites.
- Responsive Design (mobile-first approach): This standard implies adapting the design of a website depending on screen resolution from a smartphone to a computer.

5.6.2 Professional and Security Standards

The following were some of the specific standards that had to be followed in the corporate setting to ensure company data security and quality work:

- ISO 9001: It is a standard concerning quality management systems. Quality standards were set and monitored through systematic coding practices to produce good quality internship results.
- Agile Methodology: Agile standards were used by the team to manage their tasks.

5.7 Mapping with IoTWF reference model layers

IoT World Forum (IoTWF) reference model is relevant to any system where physical devices are involved with sensor-based connectivity and layered communication system. Since the present internship project involves only the aspect of web development and no IoT

device, sensor network, or communication layers, it becomes irrelevant to use IoTWF reference model for this internship. Thus, mapping this system with IoTWF layers is not applicable.

5.8 Domain Model Specification

A domain model typically refers to entities/objects that exist in a system, along with their interrelationships in that system. While the project is not related to IoT, yet a brief look into a domain model can be considered here for web-based applications.

In this project, the domain mainly consists of objects such as the User, Web Pages, and Content Elements. The user interacts with the site via browser by navigating through different web pages including Home, About, Services, etc. These web pages contain content such as images, textual data, and other navigational objects.

However, the domain model is still technology-independent and looks only at the user interactions within the website components. This makes it easier to understand the whole system structure in an easy-to-understand way.

5.9 Communication model

The communication model refers to the way information flows between different elements of the system. In this case, the most appropriate type of communication model would be the Request–Response Model.

In the Request-Response Model, the client or user makes an initial request to the server using a web browser, which then responds appropriately based on that request. For instance, if the user loads or requests a certain web page or form, the browser sends a request for the requested pages, which is processed and results in a corresponding response from the system.

The use of such a communication model is very appropriate for this particular application since web applications naturally use the request and response method. User actions like loading a web page result in making a request to the server, which then responds accordingly.

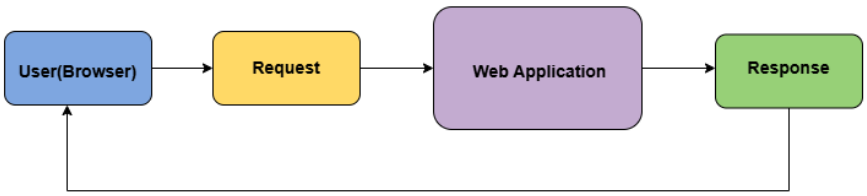

Fig5.3 Request-Response communication model in the application.

5.10 IoT deployment level

IoT deployment levels are applicable to the systems with involvement of sensors, devices, and cloud communications. As the current project relates to web applications development and does not include IoT devices and real-time sensor data exchange, IoT deployment is irrelevant to the project scope.

However, from the conceptual perspective, the system can be compared to IoT Deployment Level 1, where only one system communicates with the user, while the use of complex communication protocols or cloud IoT services is not required.

Accordingly, the project implements a very basic model of deployment when there is no need for multiple layers of IoT integration because the application functions independently or on a web platform.

5.11 Functional view

Functional view involves breaking the system down into separate functional components and specifying their role in the overall system operation. Although the current project is not IoT-related, the general approach of functional decomposition can still be used.

1.Application Layer

It is the uppermost layer of the system and serves to implement the interaction of the user

with a web application through its homepage, about page, and services page providing the necessary information.

2. Communication Layer

It provides an opportunity for two-way communication based on request/response interaction model.

3.Service Layer

This layer takes care of the features that are part of the system like page loading and navigation through JavaScript.

4. Management Layer

This layer covers aspects like file management, code management, and any changes to be made on the website.

5. Security Layer

Security aspects like input sanitization and proper handling of data are covered in this layer.

5.12 Mapping deployment level with functional blocks

The mapping of the functional blocks and the level of deployment can provide more information about the architecture of various functional blocks of the system. As the project does not use IoT, there will be a simpler approach for mapping of this kind.

The simplest form of deployment used in the project includes executing an application locally or using a web browser. The Application layer uses the interaction with the end-user by showing various web pages. The Service layer is responsible for providing functionality like navigation and interacting with users. Communication layer works with the processing of request response between the application and the browser. Management layer controls files and coding organization, while Security layer performs basic validation and executes safely.

From the described mapping, it is possible to see that all functional blocks are included into the simplest level of deployment which makes the development, maintenance, and use of the system easier.

5.13 Operational view

This view provides the details regarding the hosting, storage, and operations of the system. As the project is an application developed using the web interface, the operation is relatively easy and simple.

1.Service Hosting

Hosting options for the service include either local hosting on a system or hosting on various platforms like GitHub Pages, or other web hosting services. This makes sharing and access to the websites developed easy.

2.Storage

The project uses mostly the local storage for storing HTML, CSS, JavaScript files, and images. There will be no requirement of any external databases since the application is static.

3.Devices

The systems can be accessed through devices including laptop, desktop, tablet, or mobile phone. Responsive designing helps ensure the system can be used from any device.

4.Application Hosting

The application can be hosted in any standard web server or static hosting environment. There will be no need of any complex backend support for this application.

5.14 Other Design aspects

1.Process specification

It all starts with users interacting with the program, then comes page loading, applying

styles, processing user input, and finally showing output. All steps are performed in a linear fashion to facilitate smooth operations.

2.Information Model Specification

The program will work with information in its primitive form including text information, images, and inputs from the user. Such information is arranged in web pages using HTML elements. Information presentation rather than storage is the focus here.

3.Service Specification

Web page viewing, moving through the pages, and interaction are some of the services provided by the program. They are realized through the use of HTML, CSS, and JavaScript.

4.Additional Design Specifications

The design is focused on providing a simple, responsive, and user-friendly interface. Appropriate layout structure, styling, and validation techniques have been incorporated into the design.

Chapter 6

HARDWARE, SOFTWARE AND SIMULATION

6.1 Hardware

In this project, there are no specific hardware elements like microcontrollers, sensors, and actuators because the project is a complete web application. But the computing hardware such as laptops was utilized for designing the web application.

The development system comprises a processor that is suitable for web development processes, adequate RAM for coding and browsing, and memory to store the code and other necessary files for the project. This configuration was quite enough for using frontend technologies like HTML, CSS, Bootstrap, and JavaScript.

As there is no physical circuitry involved in the project, it becomes unnecessary to design the circuit diagram. In this case, everything takes place in the software world where the browser serves as the executing platform.

6.2 Software development tools

For the implementation of the Prola Tech and Construction web platforms, it was necessary to have special software that would be helpful in terms of coding, managing various versions of the code, and testing of the product. This software was needed in order to create high-quality code and to implement Agile collaboration within the team.

6.2.1 Visual Studio Code (VS Code)

Visual Studio Code became an Integrated Development Environment for writing and debugging frontend code.

Configuration Procedure:

The installation of the software was made on Windows-based computer equipped with at least 4 GB RAM. Extensions including "Live Server" were installed in order to provide an opportunity to see HTML5 and CSS3 changes on the go.

6.2.2 Bootstrap 5 framework:

The Bootstrap framework is the main framework for creating a responsive and "mobile-first" grid used for industrial works.

Configuration procedures:

The CDN links of Bootstrap CSS and JS bundles were included in the HTML5 code. Configured a viewport meta tag in the section of each web page to allow for responsive grid function

6.2.3 Web browsers (Google Chrome and Mozilla Firefox):

These were the main testing platforms for testing cross-browser functionality and to troubleshoot JavaScript associated functionality.

Configuration Procedures:

Utilized the "Developer Tools" (Inspect Element) to simulate screen for several devices, to make sure the CSS3 layouts are intact and the Console tab was used to monitor the execution of JavaScript for the interactive functions within the construction website.

6.3 Software code

```
<div class="col-lg-3 col-md-6">
    <div class="card">

        <!-----Service img-------->
        <div class="card-image">
            <a href="#">
                <img src="img/s3.jpg" class="card-img-top" alt="Service 3">
            </a>
        </div>

        <!----Service Content----------->
        <div class="card-body">
            <a href="#">
                <h2 class="card-title">Outdoor Living Spaces</h2>
            </a>
```

Fig 6.1 HTML Code

```
@media (max-width:991px) {
    .about-us {
        padding: 50px 0;
    }

    .about-image {
        text-align: center;
        margin-bottom: 30px;
    }

    .about-content {
        margin-left: 0;
    }

    .section-title {
        margin-bottom: 30px;
    }
```

Fig 6.2 CSS Code

```
function validateForm() {
    let name = document.getElementById("name").value.trim();
    let email = document.getElementById("email").value.trim();
    let mobile = document.getElementById("mobile").value.trim();
    let requirement = document.getElementById("requirement").value;
    let location = document.getElementById("location").value.trim();

    // Name validation
    if (name === "") {
        alert("Please enter your full name");
        return false;
    }
```

Fig 6.3 JavaScript Code

6.4 Simulation

Simulation involves analyzing how a particular system works using software to emulate the operation of that system virtually before actual real-time implementation takes place. In hardware-based projects, the main use of simulation tools involves testing circuits, microcontrollers, and system performance in general. However, in this internship project, the whole project has been centered on web development and thus there is no need for simulation of electronic circuits or any other type of embedded system.

In this case, simulation is done using web browsers. The developed websites will be tested through the web browser to test different functions and how the system behaves when accessed through browsers. It is through the simulation process that different functions such as navigation, styling, and responsiveness can be tested. Simulation in web development mainly focuses on how the system will respond to certain situations.

The developer tool in the browser has enabled us to simulate different screen sizes of various devices so as to check whether the website is responsive in all cases. Different test cases including layout checks, user interaction tests, and input verification among other test cases have been run to verify how the system responds in different scenarios.

Chapter 7

EVALUATION AND RESULTS

Evaluation plays an important role to determine if web applications created at Prola Tech are meeting functional as well as non-functional requirements. The testing procedure employed to identify, measure and validate the performance of the hardware and software units is discussed in detail in this chapter.

7.1 Test Points

Test points help in ensuring that the different components of the system are working well or not. As in this project, a web-based application will be developed, the test points will be determined mostly in terms of software functional units such as loading, styling, responsiveness, and user interactions.

Test Points in Software Functional Units:-

- HTML Unit: Testing if all the elements including headings, images, text are correctly loaded and displayed on the web page.
- CSS Unit: Verifying the correct implementation of styles in terms of color, font, spacing, and alignment.
- Bootstrap Unit: Ensuring responsiveness of the webpage on mobile, tablet, and desktop screen resolutions.
- JavaScript Unit: Validation of user input as well as functioning of interactive controls such as buttons and forms.
- Navigation Unit: Verification of redirection to appropriate pages through links and navigation menu.

7.2 Test Plan

Test plan is used to check whether each component of the web application works according to specification. The test plan describes the process of testing of different components in particular conditions.

7.2.1 Test Case:-

- Web page must load the content when visited by browser with normal conditions and valid HTML format
- CSS must apply styles if page has valid syntax and linked stylesheet
- Bootstrap must change layout depending on the screen size (mobile/tablet/desktop)
- JavaScript must validate input from the form depending on whether values are valid or not
- Navigation system must redirect the pages by clicking the link without any errors
- The form must display an error if user provides invalid data in form fields

7.2.2 Types of Testing Carried out

1. Black Box Testing

- Positive Testing: Valid inputs are provided and output results checked.

  E.g. correct data in the form = correct result

- Negative Testing: Incorrect inputs provided to see if system reacts correctly.

  E.g. empty input = error message

- Boundary Testing: Testing extreme boundaries for inputs.

  E.g. minimum/maximum number of characters in form field

2. White Box Testing

- Control Flow Testing: Checking the code flow (navigation functions, etc.)
- Data Flow Testing: Verifying proper data transfer.
- Branch Testing: Testing for conditions such as whether input is valid/invalid
- Path Testing: Testing all possible execution paths.

3. Unit Testing : The individual units like HTML pages, CSS styles, and JavaScript are checked separately to verify their accuracy.

4. Integration Testing : The units are integrated together and tested to make sure there is no problem with their interaction.

5. Software Testing : The complete web application is tested with help from web browsers.

6. System Testing : System testing ensures that all the components of the web application interact correctly without any flaws.

7. Validation

- Dynamic Validation: The application is tested by running it dynamically.
- User Requirements Validation: This validation makes sure that the website satisfies the user's requirements like responsiveness and functionality.

7.3 Test Result

The test results are used to prove if the system operates as required in various scenarios. All test cases listed in the test plan have been performed with the results documented and analyzed. Given that this is an online system, the results are based on factors such as the loading time, response of the system, validity of the input validation, and user interaction.

Table 7.1 Test Case Results

Test Case	Expected Result	Actual Result	Status
Web page loading	Page must load successfully.	Page loaded successfully.	Pass
CSS Styling	Suitable page styling.	Styles applied effectively.	Pass
Responsiveness	Page layout is device-specific.	Page layout changes effectively.	Pass
Form Validation	Error is generated for invalid input.	Input error message generated.	Pass
Navigation	Page redirect to proper page	Navigation working effectively.	Pass

7.4 Insights

During the evaluation process of the web application, many useful insights about the performance and functionality of the system have been gained. One of the significant issues that were discovered had to do with the layout consistency. In the beginning, some components were poorly positioned on mobile screens due to the wrong style and inability to respond. The problem was fixed by successfully applying the principles of Bootstrap grids and media queries.

Another insight was that the system was not doing enough to handle the input and validate data. At first, the system was taking data without proper validation, which led to the wrong or incomplete user input. This issue has been solved by applying JavaScript techniques for

checking and validating data. Minor inconsistencies in rendering styles in some browsers were another problem; they were solved by testing the application in different browsers and applying consistent coding standards.

All in all, these insights contributed to making the system more reliable and responsive. Solving such issues has greatly enhanced the performance of the website.

Chapter 8

SOCIAL, LEGAL, ETHICAL, SUSTAINABILITY AND SAFTEY ASPECTS

8.1 Social Aspects

Positive implications:

Access to information through websites makes it easier for users to comprehend what the organization has to offer. Web technologies help with communication through interfaces that are user-friendly. They also make it easier for users to learn using technology through digital education.

Negative implications:

Some people may be unable to access the website because of no internet connectivity or devices to connect with. It may also be difficult to navigate the site, making it more challenging for some users. Users might misuse the website in such a way that they provide incorrect information.

Case Study (Web Technology Impacts)

Social impact is seen positively when web technologies improve the visibility of organizations and engage the target audience. The negative impact comes in where there is misinformation and poor accessibility. Unethical data collection may also pose problems for web technology users.

8.2 Legal Aspects

Data privacy laws:

It is imperative for web applications to adhere to data protection laws to ensure that all user data is handled safely and responsibly. The Digital Personal Data Protection Act, 2023 (DPDPA), which is applicable in India, stipulates that there should be user consent, purpose

limitation, and reasonable measures for safeguarding personal data. Web applications, even the simplest ones, must ensure that there is no unnecessary storage of user data and input validation.

Rights and obligations:

The rights of users must be considered, and it is expected that any data collected from users will be properly secured and utilized only for its intended purpose. Therefore, developers and organizations must ensure that user data is safe if collected. Some basic steps such as input validation and minimal storage of data must be adhered to at all times.

Challenges:

Compliance with evolving data protection laws is one of the most significant challenges that arise in designing and developing web applications. The website must be designed to accommodate users from various jurisdictions to ensure compliance with data protection laws. There is also a need to ensure that the website avoids legal issues associated with content and images used within the web application.

8.3 Ethical Aspects

Concerning ethical issues in this project, the first important thing to note is the requirement of equality, responsibility, and low chances of harming other people during developing and using the application on the Internet. The responsibility of any developer should lie with society, because his or her task is to create websites which are easily accessed, user friendly and provide valuable information and make life easier. There are no addictive aspects in this project; it is rather informational and does not make anyone addicted to it in any way. This web application does not lead to depersonalizing individuals, since it does not make any individual do something against his or her will. There is nothing unethical about using the Internet for finding interesting facts. Ethical aspects such as originality of content, absence of misleading information, and data security were taken into consideration during developing of this application.

8.4 Sustainability Aspects

The sustainability aspects in this project are mainly associated with resource efficiency and low environmental impact. As the application is implemented online, there will be no need to use any physical material, which means the lack of resources consumption as well as absence of material waste. Using light web design, optimized images, and efficient programming methods contributes to reducing energy consumption when running the website. Using open-source tools and technologies prevents duplicate resources from being used and creates conditions for resources recycling.

The sustainability aspect is also associated with durable and scalable web development. This means that the developed website can be reused and is easy to maintain, so its modification will be possible without the necessity of redevelopment. Using efficient hosting services and minimum space for storing information minimizes server energy consumption. Speaking about social sustainability, the developed application is safe for users as it provides them with comprehensive information in a safe and easy-to-use way.

8.5 Safety Aspects

The aspects of safety in this particular project include the safety of the application itself, the reliability of the app, and its absence of risks for the users. As the project is based on the web application, the aspects of safety relate to the secure development of a web-based solution. Such safety aspects as input validation and error handling are used to make sure the application works as expected, without any issues or misuses.

Furthermore, the cybersecurity aspect can be mentioned. It will help ensure that the web-based application is protected from any potential risks or threats and that it provides reliable performance. It will also help minimize the risks associated with user interaction and make sure that all users have the opportunity to use the service without any mistakes.

Chapter 9

CONCLUSION

During the internship, I learned how to use frontend technologies such as HTML, CSS, Bootstrap, and JavaScript to create responsive websites. The steps taken during the process entailed understanding the requirement, designing the site layout, developing webpages, and testing the functionalities and responsiveness of the designed websites. The two websites created included the company website and the construction website.

From the analysis above, it is clear that the web applications are functional, responsive, and user-friendly. It meets the objectives of structuring web pages, styling properly, making responsive pages, and making the webpages interactive with JavaScript as required in the introductory part of the research paper. The tests carried out prove that the web pages are working efficiently and effectively.

There are a lot of areas where one could improve the project including using backend technologies to enable dynamic content handling and database connectivity. One could add such features as user authentication and advanced forms. Security aspects such as hashing could be incorporated to ensure the websites' safety. In addition, the design could incorporate more user-friendly user interface.

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APPENDIX – A

This appendix contains screenshots of responsive company and construction websites.

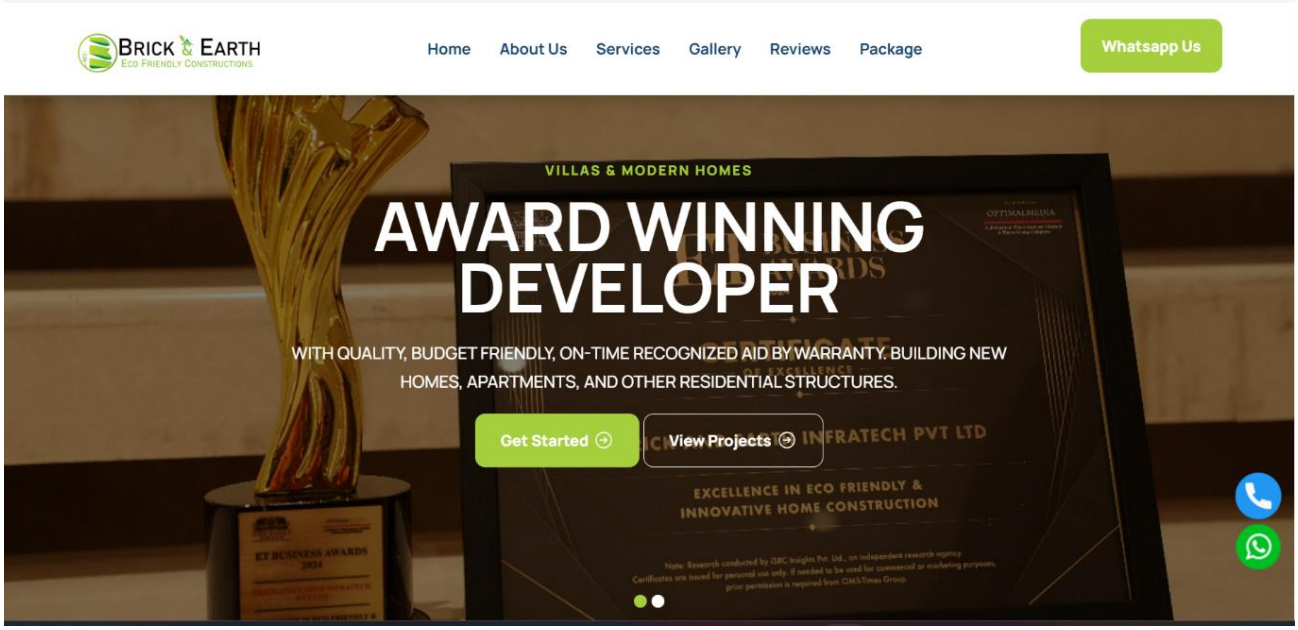

Fig A1 Construction Website(Home Page)

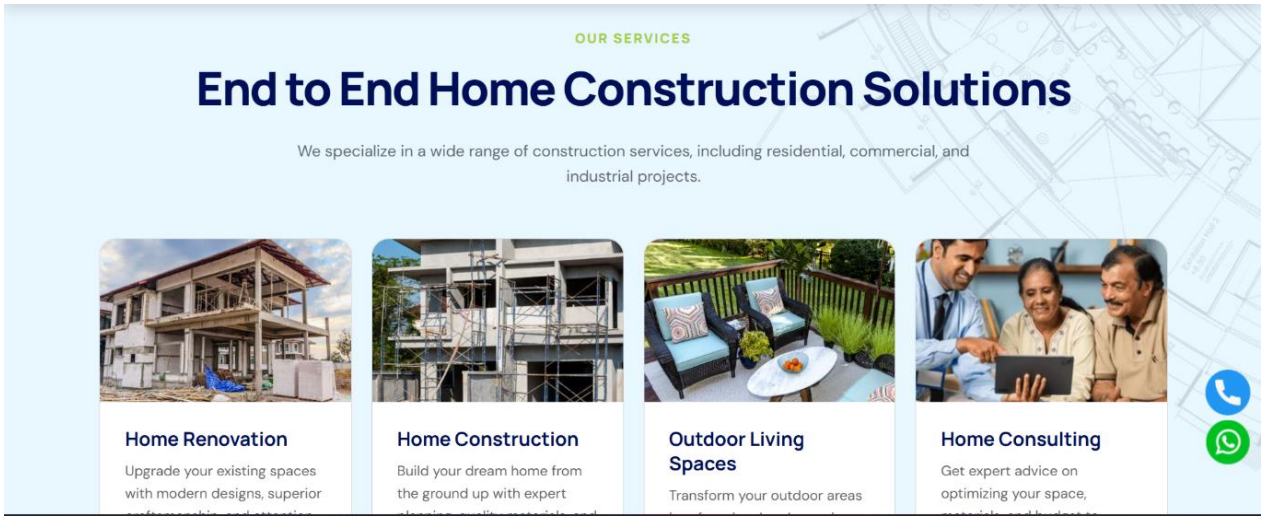

Fig A2 Service Section UI

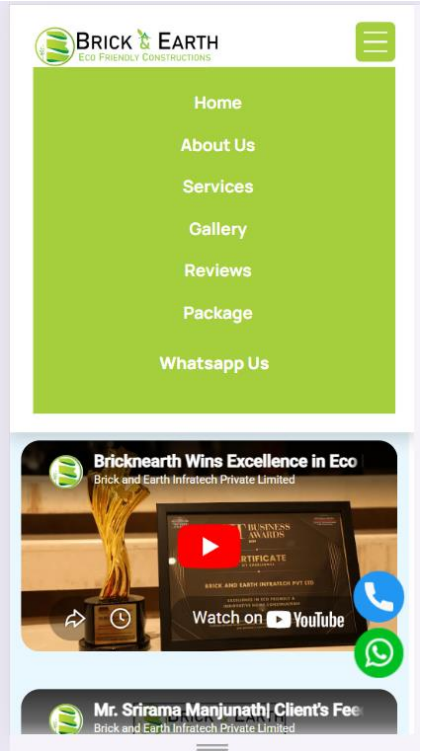

Fig A3 Mobile View Navigation

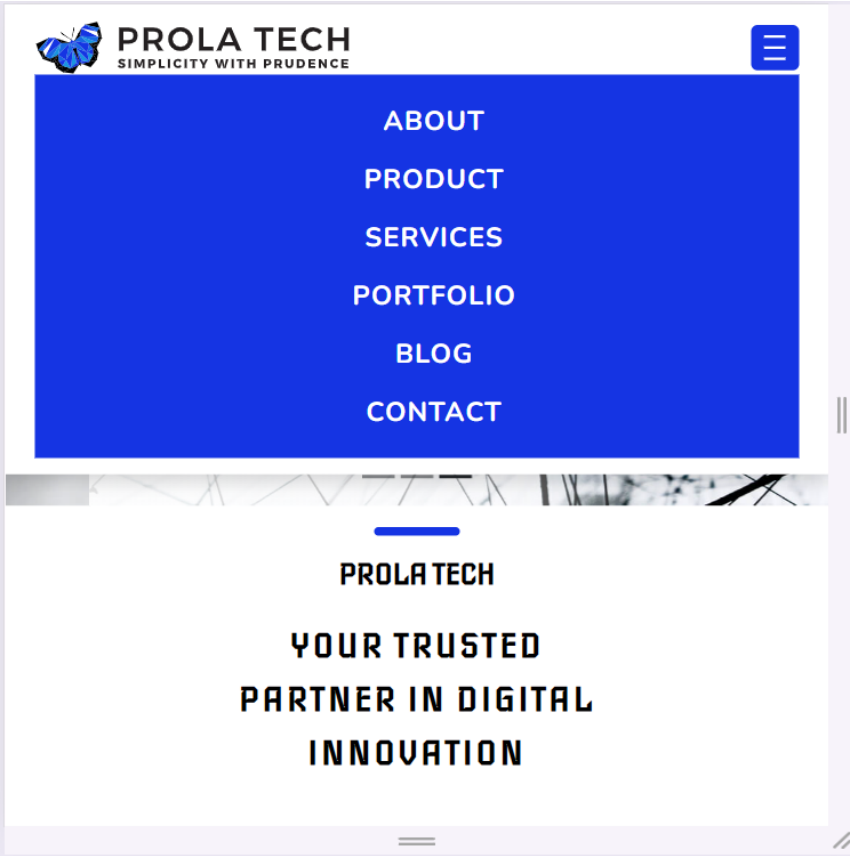

Fig A4 Company Website